

Neurolabs Android SDK Integration Guide

Technical Integration Specification

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1. Scope

This guide is for partner Android apps integrating 'neurolabs-android-sdk' directly.

2. Release History

No public API contract has been broken since v1.1.9. New options have been added with backwards-compatible defaults; integrations on v1.1.9 keep working as is. Recommended defaults: strict shelf guidance with 'validationPreset = IOS_PARITY', 'liveQualityChecksEnabled = true', 'autoCloseAfterCapture = true' for single-capture parity flows.

v1.3.2

- Mixpanel ingestion endpoint now resolves via 'NLANalyticsDefaults.MIXPANEL_ENDPOINT_EU' at SDK construction; events flow into the EU region project as expected. The standalone 'MixpanelAnalyticsTracker' class keeps the US endpoint as its default so partners reusing it for their own US-region projects are unaffected.
- 'CaptureViewModel.onFinishPressed' now claims 'isFinishing' synchronously, wraps every disk read (pending review reload, capture reload, temp-file cleanup) in 'withContext(Dispatchers.IO)', and offloads 'spillCaptureToDisk' from 'commitCapture' - the Done button stays responsive throughout multi-capture sequences and shows an inline 'CircularProgressIndicator' while persistence is in flight.
- Embedded custom-camera demo gained an in-app settings sheet (gear icon, top-right of the camera). Sliders for detection thresholds, capture min / max, live-quality FPS; switches for every boolean knob; SharedPreferences-backed persistence across launches.

v1.3.1

- Fix for an uncatchable native 'SIGSEGV' in 'com.google.ai.edge.litert.Model.nativeLoadAsset' that triggered when the TFLite asset was stored compressed in the APK, missing, or zero length. 'NLLiteRTYoloDetector' now probes the asset with 'AssetManager.openFd' first and, on failure, extracts the asset to 'noBackupFilesDir' and loads it via the file-path overload.

v1.3.0

- Relaxed guidance thresholds (perspective skew floor raised, angle-off warning relaxed) for less aggressive blocking during normal in-store motion. Android lint passes restored.

v1.2.8

- Onboarding flow: same-barcode relookup is allowed after a completed attempt so partners can retake without restarting the session.

v1.2.7

- Lint compatibility fix for product capture; no API change.

v1.2.6

- Onboarding: partner-overridable operations lookup client and capture debug exports (frame dump for offline review).
- Demo: scrollable launcher on small devices, nav-bar inset respected on the product-capture back button.

v1.2.5

- 'NLNativeCaptureConfig' / 'NLCustomCameraView' expose 'showsSequenceCounterLabel' to toggle the on-screen sequence counter.

v1.2.4

- New capture range controls: 'allowManualFinish', 'minCapturesBeforeDone', and 'maxCaptures' plumbed through 'NLNativeCaptureConfig' for bounded multi-capture flows with a Done button.
- iOS-camera parity work: tap-to-focus, lens-switcher hardening, tightened quality validation thresholds.

v1.2.3

- 'NLNativeCaptureConfig.cameraMode = AR | DEFAULT' lets partners pick between the ARKit/ARCore pipeline and the AVFoundation/CameraX 0.5/1 lens-switcher path.
- 'init' accepts 'analyticsEnabled', 'sentryEnabled', 'sentryDsn' so partner apps can opt out of telemetry or point Sentry at their own project.
- Default validation thresholds (blur, glare, perspective) tightened.

v1.2.2

- Internal: Android plugin integration handling refactor; no public API change.

v1.2.1

- iOS camera-style UI parity polish, lens-switcher state machine hardening, demo Sentry auto-init disabled.

v1.2.0

- Internal scaffolding for v1.2.x: capture range controls landed in v1.2.4, lens switcher in v1.2.4-v1.2.5, missions policy preset, retry + idempotency conformance fixture.

v1.1.9 (baseline)

- No public API contract break.
- Default task routing config uses 'taskUUID'.
- Recommended custom-camera shelf payload remains strict shelf guidance with 'validationPreset = IOS_PARITY'.
- 'liveQualityChecksEnabled' remains the key switch for pill / rotation / warning / error guidance behavior.

- 'autoCloseAfterCapture = true' remains the recommended parity flow for single accepted captures.

3. Requirements

- 'minSdk 26'
- JDK 17
- Kotlin 2.x

4. Install

```
// app/build.gradle.kts
dependencies {
    implementation(files("libs/neurolabs-android-sdk-v1.1.x.aar"))
}
```

If you consume a local 'aar', include required runtime deps (Compose/CameraX/serialization/etc.) or use the provided integration gradle script from the Cordova plugin setup.

5. SDK Initialization + Warmup

```
import ai.neurolabs.sdk.core.*
import kotlinx.coroutines.launch

class MainApp : Application() {
    override fun onCreate() {
        super.onCreate()

        val config = NLConfiguration.production(
            apiKey = BuildConfig.NL_API_KEY,
            taskUUID = BuildConfig.NL_TASK_UUID,
            debugLogging = BuildConfig.DEBUG
        )

        NeurolabsSDKCore.configure(this, config)
    }
}

// e.g. in first Activity/VM
val sdk = NeurolabsSDKCore.shared ?: return
lifecycleScope.launch {
    sdk.initialize() // the SDK's one detector, all_in_one
    sdk.waitForFullyLoaded()
}
```

The detector is downloaded, not bundled (1.7.13+). The SDK ships no model, so it adds about 6.5 MB to your app. Its one detector, 'all_in_one' (YOLO11l-Seg at 960 px, 12 classes: SKUs, 'sinc' multipacks, prices, promos, posters, gaps, every row type and 'shelf_strip'), comes from the Operations API, per org, with the operations key (or auth provider) you already configure: a signed URL, a sha256-verified, resumable, cached 55.6 MB download. The SDK prefetches it at start-up, when the credential changes and when a capture session opens. The old model names ('NLBModel.tflite', 'yolo11l_seg_sam.tflite', ...) and every 'NLYoloModelVariant' now mean 'all_in_one'.

The first launch needs network for detection. Until the model is on the device, capture works without a detector: the shutter, the queue and the light/glare/contrast checks work, 'waitForFullyLoaded()' returns, 'modelPreparationState' stays 'LOADING', detection throws the retryable 'DETECTOR_UNAVAILABLE', and the capture screen shows "Detector downloading. You can still capture." (no boxes, no "Point at a shelf", no AR anchors). When the model lands it is swapped in mid-session.

'NLDetectorModels.status(context)' gives the state and, after a failure, the failure and its code in 'reason', the same codes as iOS: 'no_credentials', 'not_entitled', 'unknown_model', 'pin_mismatch', 'sha_mismatch' (an hour before the API is asked again) and 'api_unavailable', 'download_failed' (60 s; 'compile_failed' is iOS-only), plus 'insufficient_storage' (no wait: the next capture session or 'prefetch' asks again) and 'load_failed' (a fresh copy that still will not load; not retried until the next launch). New operations credentials clear the wait. With no operations credential and no model on the device, the SDK does not fetch on its own and the status stays 'NOT_DOWNLOADED' until a credential is set (1.7.14). To have it before the first session (on Wi-Fi, after the operations credential is set):

```
lifecycleScope.launch { NLDetectorModels.prefetch(context) }
```

Keep the SDK's own download off cellular (1.7.16+). By default the SDK downloads the detector on any network. Set 'NLDetectorModels.prefetchPolicy = NLDetectorPrefetchNetworkPolicy.UNMETERED_ONLY' before 'NeurolabsSDKCore.configure(...)' and its own downloads (start-up, credential change, a waiting 'initialize', a capture session opening) wait for an unmetered network (Wi-Fi, Ethernet; a carrier's temporarily unmetered network only with Data Saver off) and start by themselves when one becomes the default network. Meanwhile 'status' is 'NOT_DOWNLOADED' with 'reason' 'waiting_for_unmetered_network', and the camera runs without a detector ("Detector unavailable. You can still capture."). An explicit 'NLDetectorModels.prefetch(context)' ignores the policy, so call it when you do want the cellular download (for example when the rep opens the camera with no detector yet).

Allow 'api.operations.neurolabs.ai' and 'storage.googleapis.com' in any egress allowlist. The detector reads its input size and class order from the model, and live frames are sized to it. It costs about 2.25 the pixels per frame of a 640-px model, so check the live frame rate on your lowest-end device first.

6. Queue Management

```
val sdk = NeurolabsSDKCore.shared ?: return

// runtime queue behavior
sdk.setAutoSyncEnabled(true)
sdk.setWifiOnlyUploadsEnabled(false)
sdk.setDeletePhotosOnUploadSuccess(true)
sdk.setMaxQueueSize(200)
sdk.setUploadRetryCount(5)

// lifecycle actions
lifecycleScope.launch {
    sdk.flushQueue()
    sdk.retryFailedUploads()
}

val status = sdk.getQueueStatus()
println("pending=${status.pendingCount} failed=${status.failedCount}")
```

Inspecting the queue at launch

After 'configure()' the queue loads the records an earlier process left on disk in the background. Until that pass ends, 'getQueuedItems()' can be empty or partial, and 'removeQueuedItem(id)' or 'clearUploadQueue()' can miss records that are still on disk. Before you inspect, withdraw or discard queued work at launch, wait for the pass:

```
import ai.neurolabs.sdk.core.awaitQueueHydrated

lifecycleScope.launch {
    sdk.awaitQueueHydrated() // the whole pass; returns at once once it has run
    sdk.getQueuedItems()
        .filter { it.sessionId == abandonedSessionId }
        .forEach { sdk.removeQueuedItem(it.id) }
}
```

- It returns once the whole pass has finished, also when the load failed (the queue then holds what it could read).
- A setter that rebuilds the queue ('setOperationsAuthApiKey', 'setOperationsBaseUrl', 'setMaxQueueSize', 'setUploadRetryCount') starts a new pass; the call waits for the queue that is current when it returns.
- On an instance that was released ('release()', or replaced by a 'configure()' with a different configuration) it throws 'IllegalStateException'. Call it on the instance 'configure()' returned.

Visits, owners and submits (1.7.15)

A host that tracks visits, or signs several users in on one device, can use these calls (see the 1.7.15 changelog entry for the full rules):

```

import ai.neurolabs.sdk.core.*

lifecycleScope.launch {
    // One queue per user and environment instead of per apiKey. Returns the new core;
    // set credentials on it. The previous owner's work stays on the device, untouched.
    val sdk = NeurolabsSDKCore.switchQueueOwner(context, config, "prod:$userId")

    // Asked once when a call is refused (409/403) because the mission's claim was lost.
    sdk.setOperationsReclaimHandler { missionId -> reclaimMission(missionId) }
    launch { sdk.submitEvents.collect { event -> /* NLSubmitEvent.Submitted / Abandoned */ } }
}

sdk.openSession(sessionId)           // the orphan sweep never submits it on its own
sdk.openOperationsSessions()         // at launch: stop or discard each open session
sdk.queuedSessions()                 // every session until its submit is confirmed
sdk.owedSubmits()                    // submits the SDK still owes
sdk.retryQueuedItems(listOf(id))     // retry chosen failed items only
sdk.discardOperationsSession(sessionId) // or discardAllOperationsWork()
}

```

- Each queued photo carries 'NLQueuedItem.missionId' (the session's mission, stamped at enqueue), 'lastErrorCode' / 'lastErrorMessage' and 'bytesSent'.
- 'clearUploadQueue()' drops queued items and notes only. To drop a visit for good, uploaded image ids and owed submits included, use 'discardOperationsSession' or 'discardAllOperationsWork'.
- The embedded camera ('NLCustomCameraView', 'NLCaptureScreen') takes 'onCapture: ((CapturedImageData) -> Unit)?', called once per photo as it joins the session; key copies by 'captureId'.

Kept photos and low storage (1.7.16)

Uploaded photos no longer take a 'maxQueueSize' slot, in any mode. A host that keeps photos after upload ('setDeletePhotosOnUploadSuccess(false)') or for a submit can bound them (see the 1.7.16 changelog entry for the full rules):

```

import ai.neurolabs.sdk.core.NLConfiguration

NLConfiguration.maxRetainedPhotos = 500           // default; a warning only, never refuses a capture
NLConfiguration.retainedPhotoExpiryDays = 14     // default null (off); photos kept for a submit never expire

```

- From 80% of 'maxRetainedPhotos' the SDK calls 'NeurolabsSDKDelegate.onRetainedPhotosHigh(retained, limit)' once per crossing. 'NLQueueStatus.retainedCount' / 'retainedBytes' report the kept photos.
- A new capture is refused with 'NLCaptureErrorCodes.STORAGE_LOW' (retryable) when the queue's storage has under 500 MB free. It reaches the host through 'onError' and the failed 'onCaptureResult', as 'QUEUE_FULL' does.

7. Open Custom Camera (Native Capture UI)

```
import ai.neurolabs.sdk.models.*
import ai.neurolabs.sdk.ui.openNativeCaptureUI

val sessionId = UUID.randomUUID().toString()

val config = NLNativeCaptureConfig(
    guidanceMode = NLCaptureGuidanceMode.STRICT,
    validationPreset = NLValidationPreset.IOS_PARITY,
    confidenceThreshold = 0.25f,
    iouThreshold = 0.45f,
    maxCaptures = 5,
    allowManualFinish = true,
    minCapturesBeforeDone = 2,
    showsCapturedCountLabel = false,
    showAlignmentGuidance = true,
    enableValidation = true,
    autoCloseAfterCapture = true,
    showDetections = false,
    showCapturedRegions = false,
    liveQualityChecksEnabled = true,
    liveQualityTargetFps = 6
)

val options = NLNativeCaptureOptions(
    type = NLShefType.SHELF,
    taskUUID = "<TASK_UUID_FOR_THIS_SESSION>",
    sendDetectionsMetadata = true,
    enablePreviewCropping = true,
    maxImageDimension = 1920,
    imageCompressionQuality = 0.85f
)

sdk.openNativeCaptureUI(
    context = this,
    sessionId = sessionId,
    config = config,
    options = options
)
```

Recommended capture payload notes:

- 'liveQualityChecksEnabled = true' is the key for pill/rotation/warning/error guidance behavior.
- Keep 'type = NLShefType.SHELF' and 'guidanceMode = NLCaptureGuidanceMode.STRICT' for full shelf guidance checks.
- Keep 'showDetections = false' and 'showCapturedRegions = false' to stay on the custom-camera guidance UI path.
- 'guidanceMode = NLCaptureGuidanceMode.GUIDANCE' should only be used as a temporary fallback if a partner specifically needs looser Android behavior while a strict-guidance regression is being fixed.

End-of-session review (1.7.16)

'finalReview = true' (on 'NLNativeCaptureConfig' or 'NLNativeCaptureOptions', options first; or 'NLCameraConfiguration.finalReview') gives the default (non-AR) camera without a multi-bay plan one review at the end instead of a per-shot preview:

- Done, or the count where the session would end today ('maxCaptures', the auto-close target, or the required count), opens "Review". A tap on a photo opens it with Crop, Delete and Previous / Next. Delete fires 'onCaptureDeleted' with source 'REVIEW'.
- Nothing is queued or uploaded before Confirm. The photos are staged on disk, so they survive the process being killed. Confirm queues the remaining photos in capture order, crops applied.
- Default 'false': every other session, and every host that does not set it, is unchanged.

8. Multi-Bay Capture (Long Shelves)

Multi-bay capture handles shelves too long for a single photo: the user side-steps along the shelf and captures one overlapping photo per bay, the SDK guides the side-step, dedups the overlap on device, and reports merged product counts across bays in a single session result.

Enable it by setting 'multiBay' on the capture config ('null' = feature off, the default):

```
import ai.neurolabs.sdk.models.*
import ai.neurolabs.sdk.multibay.NLMultiBayOptions
import ai.neurolabs.sdk.ui.openNativeCaptureUI

val config = NLNativeCaptureConfig(
    guidanceMode = NLCaptureGuidanceMode.STRICT,
    multiBay = NLMultiBayOptions(
        minBays = 2, // minimum bays before the session can finish (default
1)
        maxBays = 4, // maximum bays per session (default 8, hard cap 8)
        targetOverlapFraction = 0.30, // fraction of the previous bay kept visible (default
0.30)
        minDetectionConfidence = 0.5, // confidence floor for merge/count participation (def
ault 0.5)
        onDeviceDedup = true, // run overlap dedup at session finish (default true)
        generatePreview = true, // compose the display-only stitched preview (default
true)
        previewMaxDimension = 2048 // preview long-edge cap (default 2048, clamped 512..4
096)
    )
)

sdk.openNativeCaptureUI(
    context = this,
    sessionId = sessionId,
    config = config
)
```

The same 'multiBay' field exists on 'NLCameraConfiguration' for the custom-camera config path, and on the WebView/Cordova bridge payload (camelCase field names identical across iOS, Android, and Cordova; absent fields take the defaults above).

Reading merged results:

- Activity Result / bridge: 'NLNativeCaptureResultPayload.multiBay' ('NLNativeMultiBayResultSummary?', null unless the session ran multi-bay) carries 'countsByLabel: Map<String, Int>', 'baysCount', 'dedupTrusted', and 'previewImageFileUri'.

- Compose: 'NLCaptureScreen' / 'NLCustomCameraView' accept an 'onMultiBayResult: ((NLMultiBayResult) -> Unit)?' callback delivering the full result - 'bays: List<NLBaySummary>', 'mergedProducts: List<NLMergedProduct>' (one entry per physical product instance after cross-bay dedup), 'countsByLabel', the stitched preview ('previewImageData'), and 'dedupTrusted'.
- Use 'NLMultiBayResult.productCount' for a product tally - it sums only 'sku_single' + 'sku_multipack'; 'countsByLabel' also carries price/promo/poster buckets, so summing every label over-counts products.

Meaning of 'dedupTrusted':

- 'true' - counts and 'mergedProducts' are on-device deduped: each physical product instance is counted once across overlapping bays.
- 'false' - an adjacent-pair alignment broke, the dedup timed out, or 'onDeviceDedup' was disabled; counts are then the per-bay sum (an upper bound), not a deduped count.

Upload semantics:

- Each bay photo uploads as a normal capture through the operations queue (resumable mission flow, one image per capture).
- On 'stopSession' the mission submit attaches a compact merge summary ('{bays, counts_by_label, dedup_trusted, ref_homographies, sdk_version}') as structured 'client_metadata.device_dedup' plus, transitionally, a 'notes' duplicate for older backends - so the backend can re-dedup authoritatively.
- SDK-managed capture flows record that summary automatically. Only hosts presenting their own custom capture UI must pass 'NLMultiBayResult.submitNotesJSON()' to 'sdk.setMultiBaySubmitNotes(notes, sessionId)' before 'stopSession'.

Config-only mode:

- There is no partner-facing in-camera mode toggle. The Single Multi-bay capture-mode chip and sheet are internal demo/debug chrome gated behind 'NLSDKDebug.internalCameraToolsEnabled' (default 'false'; never enable it in partner builds). Partners control the mode purely via configuration: 'multiBay' set = multi-bay session, 'multiBay = null' = single-bay session - programmatic 'multiBay' configuration is unaffected by the gate.
- The Neurolabs demo app enables that chrome only in local debug builds; Firebase/release demo builds default to single-bay exactly like partner builds. The multi-bay feature itself is fully available in all builds via configuration - only the demo defaults changed.

9. Post-Processing Hooks

Use Activity Result API and process captures when camera returns.

```

private val captureLauncher = registerForActivityResult(
    NLCaptureActivity.Contract()
) { result ->
    when (result) {
        is NLCaptureActivityResult.Success -> {
            // capture metadata is in result.payload
            val count = result.payload.captures.size
            Log.d("NLB", "camera returned with $count captures")
        }
        is NLCaptureActivityResult.Cancelled -> {
            Log.d("NLB", "camera cancelled")
        }
        is NLCaptureActivityResult.Error -> {
            Log.e("NLB", "camera error: ${result.error.message}")
        }
    }
}
}

```

10. Delegate/Event Callbacks

```

sdk.delegate = object : NeurolabsSDKDelegate {
    override fun onCaptureResult(result: NLNativeCaptureResult, captureId: String, sessionId: String) {}
    override fun onError(error: NLError, sessionId: String?, messageId: String?) {}
    override fun onQueueStatusChange(status: NLQueueStatus) {}
    override fun onLoadingProgress(progress: NLLoadingProgress) {}
    override fun onQueueItemEnqueued(item: NLQueuedItem) {}
    override fun onUploadSucceeded(item: NLQueuedItem) {}
    override fun onUploadFailed(item: NLQueuedItem, error: NLError) {}
}

```

11. On-Device Recognition (v1.7.1)

Per-account, config-gated: your operations API key resolves YOUR org's recognition config and vector dataset - enabling an account is a backend flip, no app change. Every stage degrades gracefully; capture is never blocked by recognition.

```

import ai.neurolabs.sdk.recognition.NLRecognitionBootstrap

val bootstrap = NLRecognitionBootstrap(
    context,
    NLRecognitionBootstrap.Configuration(
        operationsApiKey = "<your operations key>",
        operationsBaseUrl = "https://api.operations.neurolabs.ai/v1",
    ),
)
bootstrap.start(sdk, applicationScope) // safe to call again; see below
// bootstrap.status (StateFlow) "provider: ready (N items)" etc.
// bootstrap.statusSummary also says when the chain was built offline

```

Offline. Every build that completes online saves the recognition config and the snapshot's signed manifest for that operations key (the file is named by a one-way fingerprint of the key; the key is never stored). When a later build cannot reach the Operations API (no signal, a timeout, a captive portal, a 5xx), it builds from that record for up to 7 days after the last full sync ('bootstrap.offlineCacheTTL'), as long as the embedder and every snapshot shard are still on the device and pass their sha256 checks. 'bootstrap.offlineSnapshotDate' is then the time of that sync, and 'statusSummary' reads 'provider: ready (offline, 42 items, snapshot from 2026-10-01)'. A 401 or 403 still switches recognition off and forgets the record.

Short-lived tokens. If your bearer is a session token rather than a fixed key, build the configuration with your 'NLOperationsAuthProvider' and a stable identity for the rep or org instead:

```
NLRecognitionBootstrap.Configuration(  
    authProvider = myAuthProvider, // e.g. NLRefreshableAuthProvider { exchangeSessionToken()  
}  
    offlineIdentity = repId,          // stable across tokens; must not be blank  
    operationsBaseUrl = "https://api.operations.neurolabs.ai/v1",  
)
```

Every request takes the provider's current token. A 401 drops it and the request is made once more with a fresh one; only a second 401 switches recognition off and forgets the record. The offline record is named after a one-way hash of 'offlineIdentity' (never stored as is), so a record saved under yesterday's token serves today's, and a cold start with no network (and so no token) still builds from it. A provider that cannot give a token counts as offline. A blank identity throws 'IllegalArgumentException'. 'copy()' on this configuration returns an API-key configuration: build a new one instead.

Retries. A build that ended unavailable is not final: calling 'start' again builds again once 60 s have passed after a network-type failure, or 1 h after a definitive one (disabled, refused, mismatched). After a network-type failure the bootstrap also rebuilds by itself when the device's network comes back (at most 5 times per instance). A later ready build injects its provider into the SDK like the first.

The embedder '.tflite' (DINOv3 ConvNeXt-Tiny) downloads through the Operations API ('GET /v1/device-models/dinov3-convnext-tiny-android-tflite-v1', sha256 and size pinned in the SDK) once recognition-config has enabled recognition for your org, and is cached, so later launches need no network. It is the only embedder the SDK runs: a model your app ships as an asset is never used. When the model cannot be had, recognition is off with the reason, for example 'embedder unavailable (not_entitled)'. The release AAR carries the Qdrant Edge native libraries; the default index remains the built-in brute-force this release.

Tap-to-product-card: captures expose per-detection matches via 'sdk.recognitionMatches(forCaptureId:)' pair the SDK's 'DetectionOverlay(onDetectionTap = ...)' with 'NLDetectionInspectorSheet(detection, match, resolver, onDismiss)' and an 'NLProductDetailsResolver' built over 'NLOperationsCatalogClient'. Offline or unresolved products fall back to a UUID + similarity row.

12. Notes

- Per-session upload routing is supported with 'options.taskUUID'.
- 'autoCloseAfterCapture=true' closes after Save from preview/review flow.
- Use 'allowManualFinish = true' with 'minCapturesBeforeDone' and 'maxCaptures' when you want "at least X, up to Y" capture sequences.
- Set 'showsCapturedCountLabel = false' to hide the 'N captured' chip in the top bar.
- For partner apps, avoid passing base64 images across process boundaries.